Problem D. Discrete Fourier Transform

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	512 megabytes

Given a sequence of integer $f_0, f_1, \ldots, f_{n-1}$, the discrete Fourier transform gives a sequence of complex numbers $F_0, F_1, \ldots, F_{n-1}$ that

$$F_t = \sum_{s=0}^{n-1} f_s \; e^{-2\pi ist/n}$$

for each $t = 0, 1, \ldots, n-1$, where $e^{i\theta} = \cos\theta + i\sin\theta$, and i is the imaginary unit that $i^2 = -1$.

You may reset f_k to any integer value to minimize the maximum value among $|F_0|, |F_1|, \ldots, |F_{n-1}|$, where $|z| = |p + qi| = \sqrt{p^2 + q^2}$ $(p, q \in \mathbb{R})$ is the modulus of the complex number z.

Input

The first line contains two integers n $(1 \le n \le 2\,000)$ and k $(0 \le k < n)$.

The second line contains n integers $f_0, f_1, \ldots, f_{n-1}$ $(-2\,000 \le f_i \le 2\,000)$.

Output

Output a line containing a single real number, indicating the minimum of the maximum value among $|F_0|, |F_1|, \ldots, |F_{n-1}|$ after resetting f_k to any integer value.

Your answer is acceptable if its absolute or relative error does not exceed 10^{-9}. Formally speaking, suppose that your output is a and the jury's answer is b, your output is accepted if and only if $\frac{|a-b|}{\max\{1,|b|\}} \le 10^{-9}$.

Example

standard input	standard output
3 2 1 1 0	2.0